Performance adjustments by modifications and rework of the pump on test

1. Purpose

This document describes a consistent process for modifying the centrifugal pump characteristics by under-filing, over-filing or blunting.

An accurate guidance on the expected performances modification using these processes is provided.

2. Scope

This procedure is used to determine the expected pump performances by applying under-filing, over-filing or blunting of the impeller. This approach should be used as last resort to adjust the pump performance curve to the customer's operative conditions whenever no other mean can be used to this scope.

All these methods could provide big scatter from the expected outcome. This is mainly caused by variances on the impeller casting or human errors during the machining phase.

The procedure is applicable to all centrifugal pumps for the Nova Milanese and La Spezia plants.

3. Definitions

a_2	impeller outlet width
PS	impeller pressure side
SS	impeller suction side
t_2	thickness of impeller blade near TE
TE	impeller trailing edge
β_{2B}	impeller outlet blade angle

4. General Procedure - Description

Under- filing, over filing and blunting are methods based on the removal of material near the impeller TE by hand dressing. This modification provides a change of the outlet flow field of the impeller, producing an alteration of the pump hydraulic performances.

4.1 Under-filing

Under filing is the removal of material on the suction side of the impeller TE, as shown in the figure below. This approach should be used whenever a gain of head is required to reach the rated performances. The increase of head is achieved by the enlargement of the discharge area a2 in the impeller vanes and by the change of impeller outlet blade angle β2b.

Figure 1 scheme of the impeller under-filing [2]

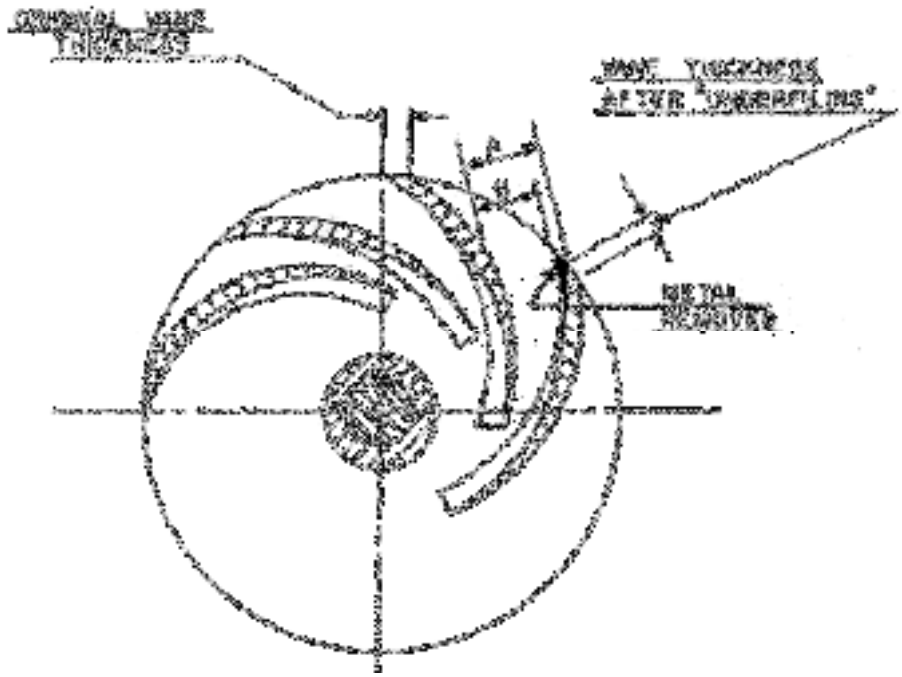

Figure 2 scheme of the impeller under-filing [3]

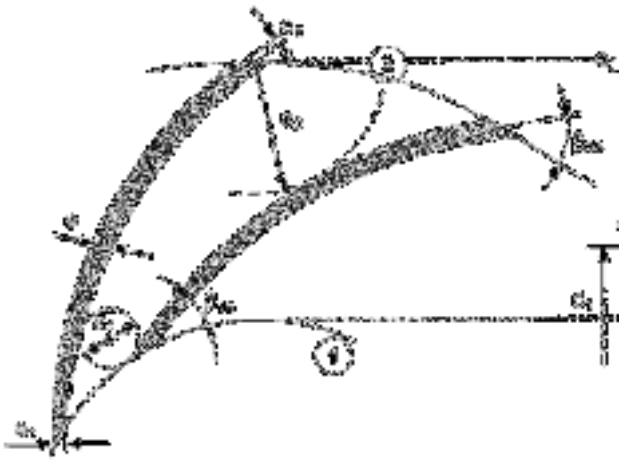

Figure 3 impeller geometric parameters [1]

4.2 Over-filing

Over filing is the removal of material in the pressure side of the impeller blade near the TE. It is used to decrease the head of the pump by reducing the impeller outlet beta angle β2b. The scheme of the over-filing is shown in Figure 4

Figure 4 over-filing scheme [2]

4.3 Blunting

Blunting or square discharge vane tip is the removal of material on a corner of the impeller blade near to the TE as shown in Figure 5. This method will provide increase of head due to the change of the impeller flow discharge angle.

Figure 5. blunting scheme [2]

4.4 TE rounding

Rounding the TE consist of a realization of a rounded shape on the outlet of the blades.

This method generally produces a gain of head and efficiency due to the reduction of losses associated to the interaction between the wake of the impeller and the casing. The rounding of the TE is usually coupled with under-filing.

A scheme of the TE rounding is shown in Figure 6

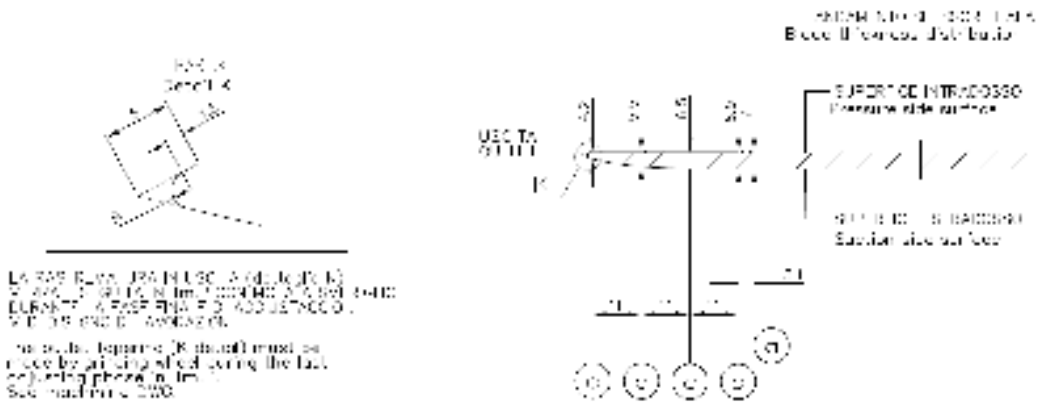

Figure 6. scheme of TE rounding

5. Performance Determination Process

5.1 Under-filing

Determine the effect of under-filing by calculating the increase in a2 (also called a2'), that the under-filing will achieve. (See also Figure 3).

Use chart on Figure 7 to determine the expected range of head increase based on the ratio of a2'/a2

Typically the highest gain will be achieved with low blade outlet angles. Impellers with blade angles of =<21° will tend to the upper limit, impellers with blade angles >=26° will tend to the lower limit.

This values refers to the best efficiency point operative condition.

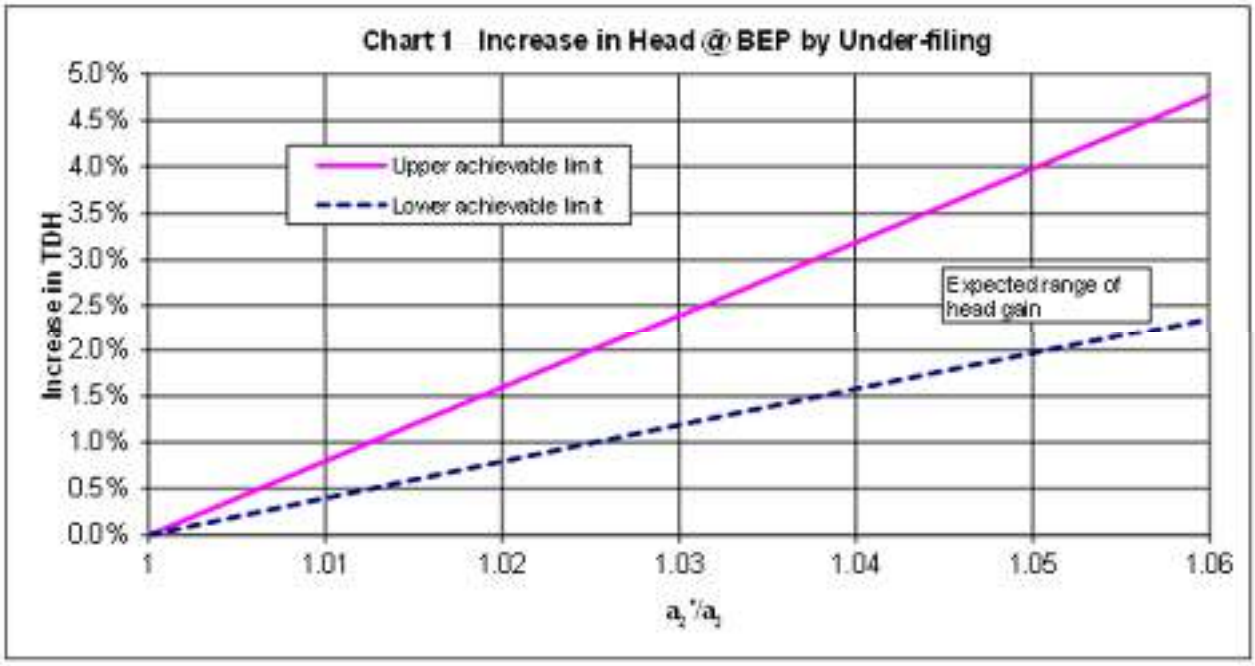

Figure 7 expected head increase by under-filing

A schematic representation of the modification of the performance curve of pump by under-filing the impeller blades is proposed by [1] in Figure 8.

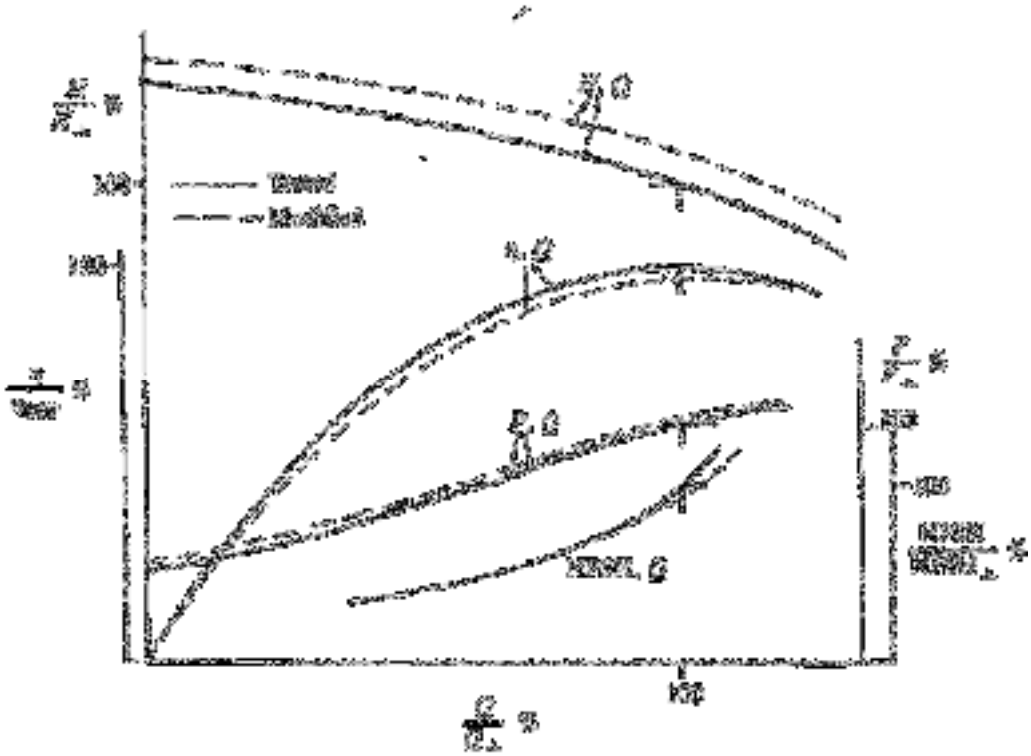

Figure 8 modification of performance curve by under-filing the impeller blades [2]

5.2 Over filing

The prediction of the performance modifications by over-filing the impeller blades is difficult due to several geometric aspects (i.e. blade angle distribution, specific speed). For this reason, no criterion will be suggested.

A proper over-filing layout that meets the required results will be achieved through the measure of the tests for several intermediate over-filings.

Head at shutoff usually remain unchanged during over-filing so this method should be used only if a steeper curve is needed, but better results can be achieved by trimming the impeller external diameter, also using oblique trim

Neumann [2] suggest a scheme of the head curve modification to evaluate the head reduction, as shown in Figure 9.

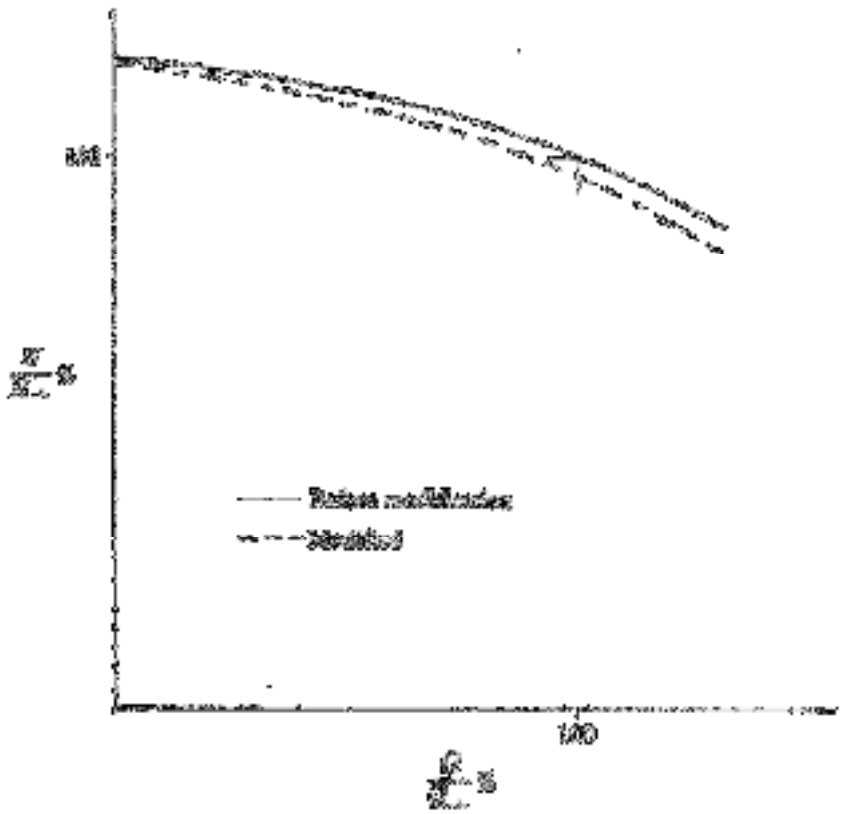

Figure 9 effect of over-filing the impeller vanes – Q-H curve [3]

5.2 Blunting

Determine the effect of blunting by the final t2 value after blunting and reading off the TDH increase from Figure 10.

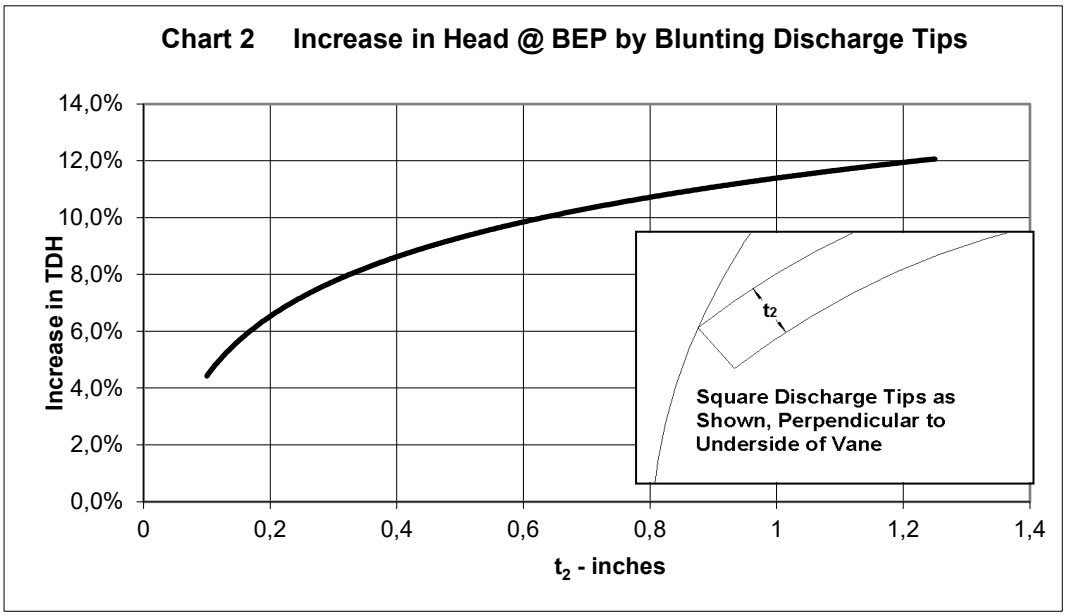

Figure 10. chart of the head increase due to blunting

A schematic representation of the modification of the performance curve of pump by blunting the impeller blades is shown by [1] in Figure 11

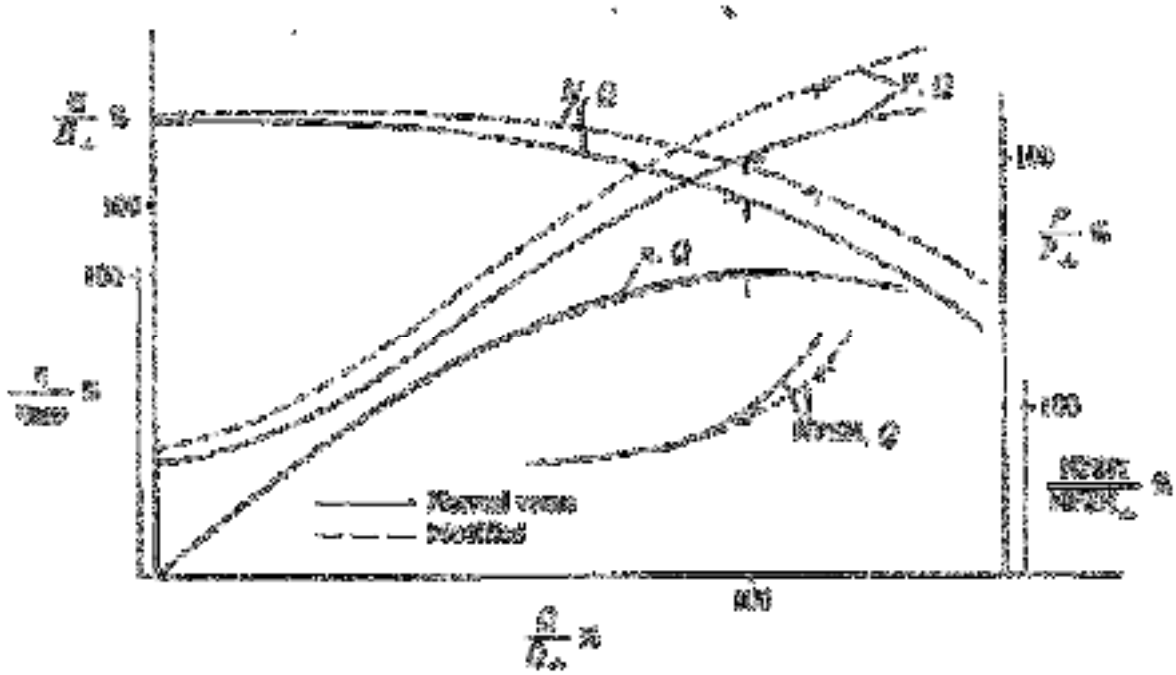

Figure 11. performance modification by blunting the impeller [2]

5.3 TE rounding

Rounding the impeller TE improves the head of the stage around up to +3%. The scatter depends on the shape of the impeller vanes, the thickness, the outlet blade angle. The efficiency is slightly improved up to +0.5% max.

6. Limitations

Doing under-filing it is recommended that the minimum vane thickness shall not be less than 30% of the original vane or 0.18 inches whichever is greater. A scheme of typical under-filing, suggested by [4], is shown in Figure 12

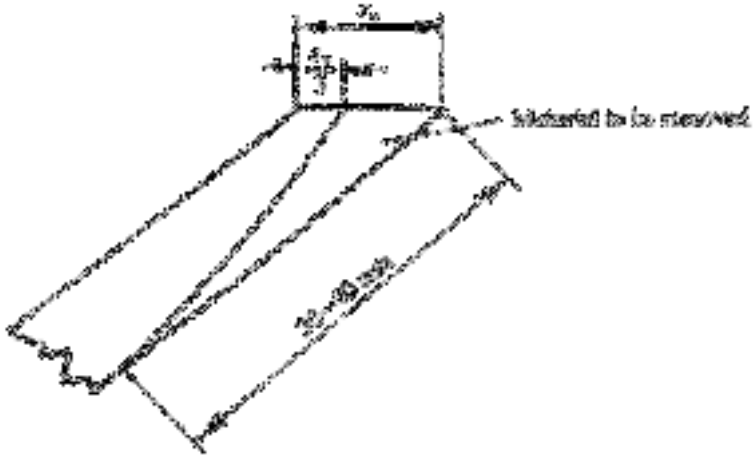

Figure 12. typical shape of under-filing [2]

It has been found that sometimes under-filing negatively influences pump efficiency (although it can also influence positively as well). The effect is not so large but it can be expected to be in the range from +0.3% to -0.3%.

Under-filing will cause the shutoff head to increase by approximately 50% of the head increase achieved at BEP. Thus it has the effect of flattening the HQ curve.

Over-filing the impeller gives uncertain results on the performance curve so it should be used only if a steeper curve is needed. By the way, the best method to reduce the head is by trimming the external diameter, using also oblique trim, so this approach should be used only if trimming of external diameter can't be applied.

Blunting appears to a neutral to positive effect on efficiency in the range of +0% to +0.5%, although no conclusive data exists.

Blunting appears to affect slightly the head at shutoff as shown by [2]

A combination of blunting, under-filing or TE rounding is possible but the overcome doesn't correspond to the sum of each contribute and must be carefully evaluated through test.

TE rounding is usually associated to under-filing. It is recommended that the minimum vane thickness shall not be less than 25% of the original vane or 0.18 inches whichever is greater

7. References

[1] Centrifugal Pumps, Fourth Edition by Johann Friedrich Gulich, Springer Nature Switzerland AG 2020

[2] Neumann, The interaction between geometry and performance of a centrifugal pump, ed.1991

[3] Lobanoff Ross – Centrifugal pumps design & application, 2nd edition